

Ahmad Saad

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UAE Golden Visa holder

PROFESSIONAL EXPERIENCE

Samsung Gulf Electronics

Dubai, United Arab Emirates

AI/ML Engineer

Jun. 2026 - Aug. 2026

- **Architected and solely built** a production trend intelligence platform (TypeScript, Node.js, PostgreSQL) on the official TikTok API for Business alongside commercial data vendors, measuring rising sounds, formats, and topics across 8 discovery lanes and publishing an evidence-grounded dashboard for the regional marketing team.
- **Designed a measurement-first detection engine** where deterministic signals (caption n-gram clustering, creator spread, adoption curves) decide verdicts and LLMs only describe them, integrating **Anthropic Claude** for brand-fit judgment and vision perception across Arabic and English content, with model agreement validated in code.
- **Owned the build end-to-end as a forward deployed engineer**, meeting directly with marketing, creative, and innovation stakeholders to scope requirements and iterate the tooling in short feedback cycles.
- **Engineered self-calibrating cost governance** across paid data and inference APIs with per-run spend meters, a cross-process daily budget ledger in PostgreSQL, and balance-gated runs that hold combined data and inference spend to a fixed single-digit-dollar daily budget.
- **Shipped and operated the full stack**: a Dockerized collector running 5 scheduled pipelines, a Next.js dashboard deployed to Cloudflare Workers behind Cloudflare Access, and custom edge middleware for per-IP rate limiting and CSP security headers.
- Introduced internal social media trend research with the in-house platform, **reducing an estimated \$60K in spend** compared to similar platforms, and supported regional launch campaigns with its trend signals, contributing to a projected **20% increase in engagement rates** and earning recognition from regional HQ.

American University of Sharjah

Sharjah, United Arab Emirates

Undergraduate Research Assistant, Computer Science and Engineering

Oct. 2024 - Jul. 2025

- **Built a speech emotion recognition pipeline** with a faculty supervisor, classifying emotion from audio snippets using log-mel spectrogram representations.
- Fine-tuned **ResNet** and **EfficientNet** backbones in **TensorFlow** and **Scikit-Learn**, reaching roughly **75% accuracy** after augmentation and hyperparameter tuning.

Center of Innovation in Teaching and Learning, AUS

Sharjah, United Arab Emirates

AI Hub Assistant

Nov. 2024 - Present

- **Led the HTML and CSS redesign** of the university's AI website against stakeholder requirements, prototyping in Figma and improving accessibility and navigation.
- **Researched emerging AI tooling** and produced written evaluations that shaped how faculty adopted AI in teaching, alongside ongoing maintenance of the live site.

Chief Nest

Remote (Riyadh, Saudi Arabia)

Account Manager

Sep. 2025 - Present

- **Directed a company-wide GTM strategy** for a client, scaling operations and restructuring revenue streams to lift sales performance.
- Rebuilt client workflows across process management tools, improving task visibility and cutting operational overhead.

ONBRND

Remote (California, USA)

Operations Associate

Sep. 2025 - Feb. 2026

- Restructured client workflows in ClickUp, **reducing operational inefficiencies by 25%**, and supported the CEO with decision-ready operating insights.

Electro-Mechanical Company LLC

Abu Dhabi, United Arab Emirates

Engineering Intern

Dec. 2025 - Jan. 2026

- Supported **SCADA** and substation automation projects across Siemens medium-voltage switchgear platforms (NXAIR, NXPLUS C, 8DJEH), covering EPC execution workflows, technical submittals, procurement, and on-site material inspections for large-scale water infrastructure.

PROJECTS

Trend Radar, Production TikTok Trend-Detection Platform

github.com/ahm-adsaad/trend-radar

TypeScript, Node.js, PostgreSQL, Docker, Next.js, Anthropic Claude API. Sanitized public extract of a platform shipped to production.

- **Engineered the collection layer for reliability** with cursor-paged retrieval, retry and exponential backoff, scheduled lanes made idempotent per UTC day so retries and redeloys cannot double-bill, and stale-but-labeled fallbacks that keep the dashboard truthful when a vendor degrades.

- **Enforced a strict model boundary** with Claude Sonnet for judgment and Haiku for vision, so deterministic measurements decide every verdict and each published figure carries provenance back to its source signal.
- Maintained quality with **315 automated tests** and **16 versioned SQL migrations** across a monorepo with CI on every push.

LocalAI, On-Device RAG Document Q&A

github.com/ahm-adsaad/LocalAI

TypeScript, React, WebGPU, WebLLM, Transformers.js, IndexedDB.

- **Built a privacy-preserving RAG system** that runs PDF ingestion, embedding, retrieval, and generation entirely in the browser, with no backend and no document data leaving the device.
- **Implemented the full retrieval pipeline**: layout-aware PDF extraction, overlapping chunking, **MiniLM sentence embeddings** (q8), and **hybrid retrieval** combining dense cosine similarity, BM25, and Reciprocal Rank Fusion.
- **Ran streaming on-device generation** with Qwen 2.5, Phi-4, and Llama 3.2 (Q4f16) on **WebGPU**, hosting WebLLM and Transformers.js in a Web Worker so model load, embedding, and token decode never block the UI, with a hard no-remote-inference guarantee.
- Improved retrieval quality on weak overview queries and noisy dashboard PDFs through query expansion, prose-density reranking, and intro-chunk pinning, then tuned context window, max tokens, and excerpt length against browser prefill and decode latency.
- Persisted documents and embeddings in **IndexedDB** with offline weight caching, and added a VRAM-aware multi-model catalog so users switch small language models without a server.

Mano's Basic Computer Simulator

github.com/ahm-adsaad/manos-basic-computer-simulator

- **Developed a cycle-accurate CPU simulator** of Mano's Basic Computer with full datapath, complete ISA support, and micro-operation level control, plus a Python CLI debugger for stepwise execution and state inspection.

Energy Harvesting LoRaWAN Sensor Node

- **Designed and fabricated a 2-layer custom PCB** integrating an ATMEGA4809 MCU, XL1276 LoRa transceiver, AEM10330 PMIC, and BME680 sensors, covering schematic capture, IPC-2221-compliant routing, and inline PPK2 headers for current profiling across the LoRaWAN transmission cycle.

TECHNICAL SKILLS

- **Programming**: Python, TypeScript, C++, SQL, JavaScript, MATLAB, HTML/CSS
- **AI and ML**: PyTorch, TensorFlow, Scikit-Learn, Transformers.js, RAG, embeddings, vector search, hybrid search (BM25, cosine, Reciprocal Rank Fusion), prompt engineering, LLM integration (Anthropic Claude API), computer vision, model fine-tuning, quantization (Q4, q8), on-device and edge inference (WebGPU, WebLLM)
- **Data and Systems**: PostgreSQL, Supabase, Pandas, NumPy, Matplotlib, Seaborn, data pipelines, ETL, REST API integration, rate limiting, API cost optimization, database migrations, automated testing
- **Tools and Platforms**: Claude Code, Cursor, Docker, Node.js, Next.js, React, Git, CI/CD, Cloudflare Workers, Railway, Figma, EasyEDA
- **Languages**: English (bilingual proficiency), Arabic (bilingual proficiency), Italian (limited working proficiency)

EDUCATION

American University of Sharjah

Sharjah, United Arab Emirates

Bachelor of Science, Computer Engineering

Aug. 2023 - Dec. 2026

- **GPA: 3.96 / 4.0**. Tau Beta Pi (Chapter President), Academic Merit Scholarship, Dean's List (x6), Chancellor's List (x3).
- **Relevant coursework**: Machine Learning, Data Structures and Algorithms, Computer Architecture, Computer Networks, Wireless Communications.